PHYS 104 – PHYSICS II

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Lecture 2

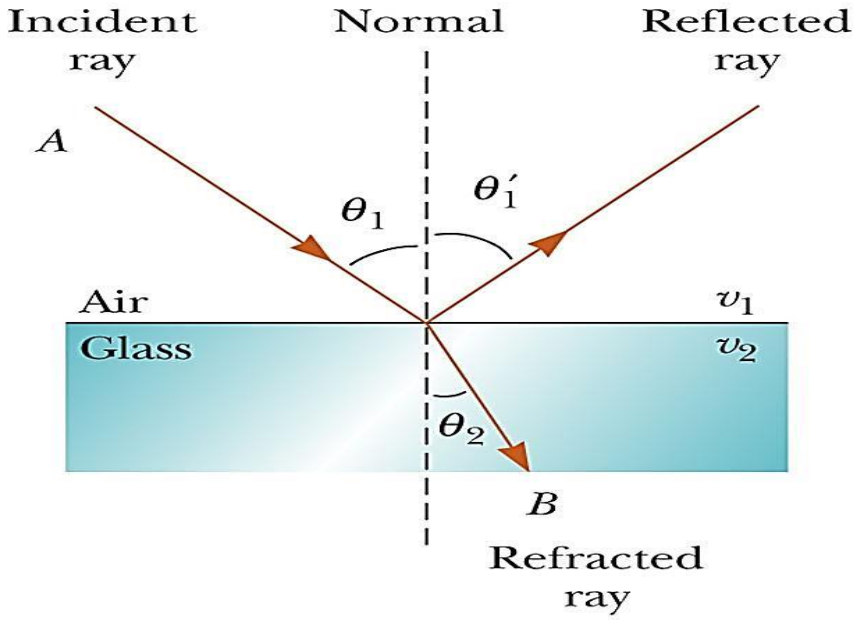


Chapter 35

The Nature of Light and the Principles of Ray Optics

Refraction of Light

- It is the change of direction of light as it passes from one medium to another.
- When a ray of light traveling through a transparent medium encounters a boundary leading into another transparent medium, part of the energy is reflected and part enters the second medium.
- Angle of refraction: angle between refracted ray and normal to surface
- Refraction rules (laws):



$$\textbf{Rule I}: \quad \frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}$$

v_1: speed of light in 1st medium
v_2: speed of light in 2nd medium

Rule II: ***The incident light ray, refracted ray, reflected light ray and the normal to reflecting surface must all lie in the same plane***

Angle of deviation: Angle between the refracted ray and extension of the incident ray

$$\boldsymbol{\beta = |\theta_1 - \theta_2|}$$

Refractive index (index of refraction) (n): It is the ratio of the speed of light in air (vacuum) $\left(c = 3 \times 10^8 \frac{\text{m}}{\text{sec}}\right)$ to the speed of light in medium.

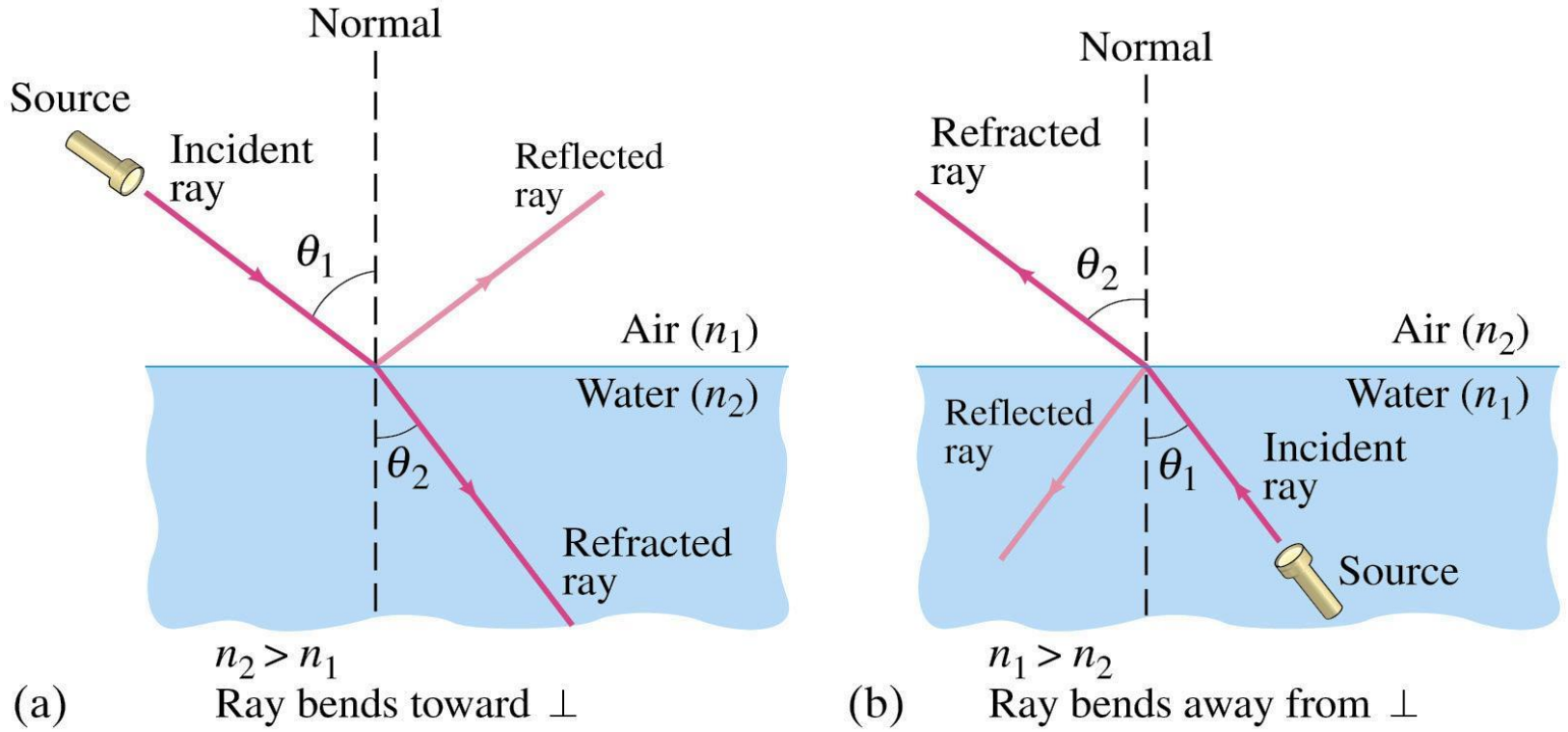
$$\boldsymbol{n = \frac{c}{v}} \quad \boldsymbol{unitless} \qquad \boldsymbol{n \propto \frac{1}{v}}$$

$$n = 1 \; for \; air \quad \& \quad n > 1 \; for \; any \; other \; medium$$

$$n_w = \frac{c}{v_w} \qquad n_g = \frac{c}{v_g} \qquad n_{oil} = \frac{c}{v_{oil}}$$

Relative refractive index between two mediums: the ratio between the refractive indices of the two mediums

$$_1n_2 = \frac{n_2}{n_1}$$



$v_2 < v_1$

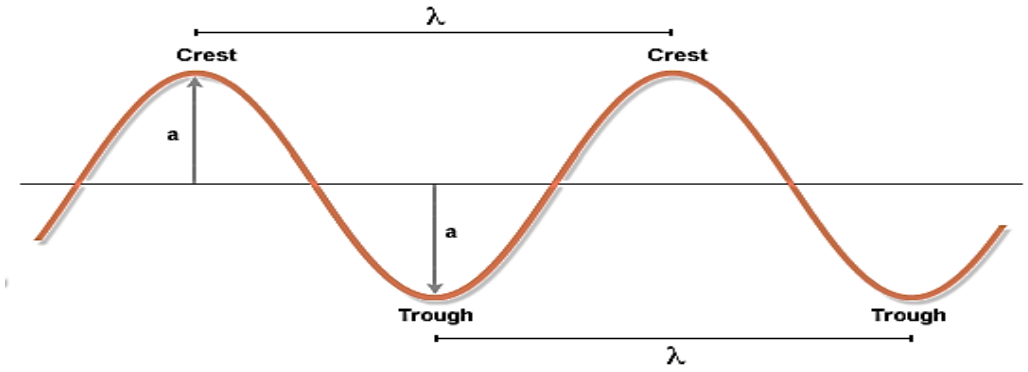
$v_2 > v_1$

Snell's law:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

- For any wave:

$$v = \lambda f$$



v: $speed\ of\ wave$

λ: $wavelength\ (distance\ between\ two\ successive\ crests\ or\ troughs)$

f: $frequency\ (number\ of\ complete\ cycles\ in\ one\ sec)$

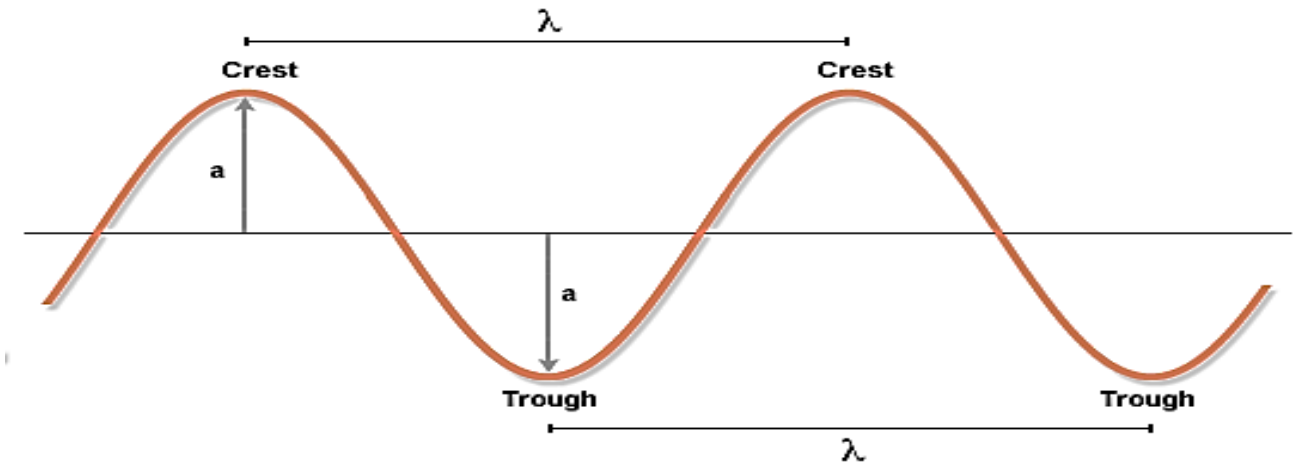
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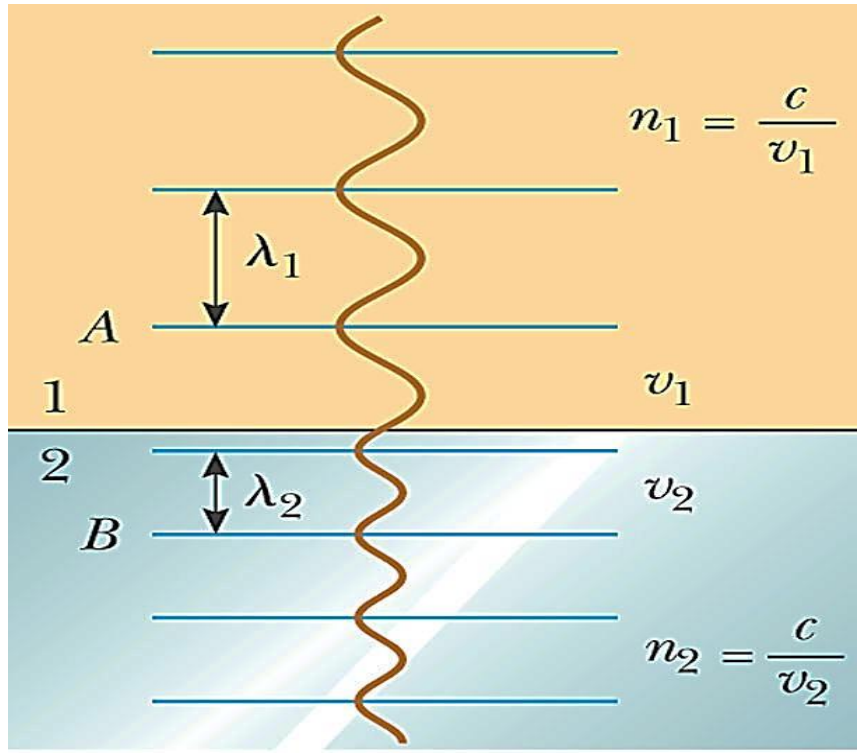
- When the light passes from one medium to another, the speed v and the wavelength λ change while the **frequency f does not change.**

$$v_1 = \lambda_1 f \quad \& \quad v_2 = \lambda_2 f$$

$$\boxed{\frac{n_1}{n_2} = \frac{v_2}{v_1} = \frac{\lambda_2}{\lambda_1}}$$

$$n \propto \frac{1}{\lambda}$$

$$\boxed{\lambda_m = \frac{\lambda_{air}}{n_m}}$$



Example 2:

A beam of light of wavelength 550 nm traveling in air is incident on a slab of transparent material, the incident beam makes an angle of 40° with the normal, and the refracted beam makes an angle of 26° with the normal. Find the index of refraction of the material.

Solution

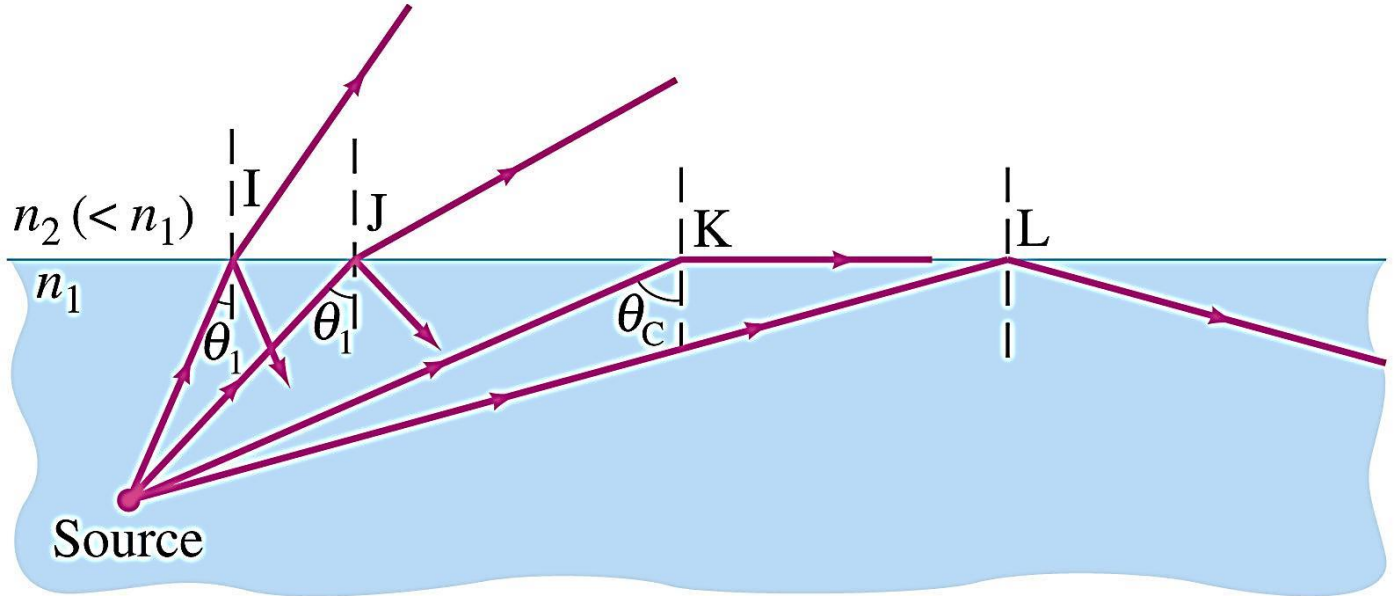
$$n_1 = 1 \qquad \theta_1 = 40° \qquad \theta_2 = 26°$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$n_2 = \frac{n_1 \sin\theta_1}{\sin\theta_2} = \frac{1 \times \sin 40}{\sin 26} = 1.47$$

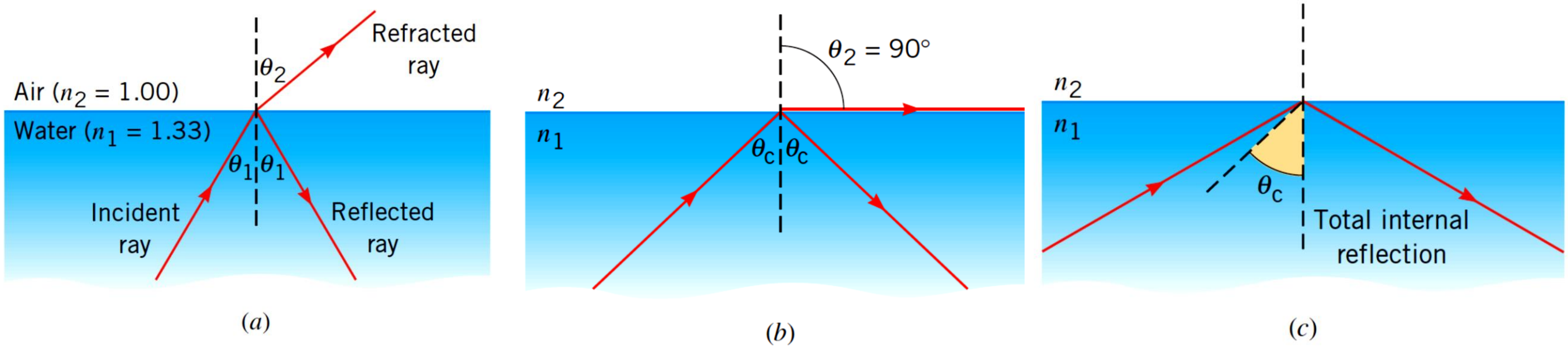
Total Internal Reflection of Light

- When light passes from a medium of larger refractive index into one of smaller refractive index (for example, from water to air) the refracted ray bends *away* from the normal.
- As the angle of incidence increases, the angle of refraction also increases.
- When the angle of incidence reaches a certain value, called the critical angle $\boldsymbol{\theta_C}$, the angle of refraction is $90°$. Then the refracted ray points along the surface.
- When the angle of incidence exceeds the critical angle, as in Figure c, there is no refracted light. All the incident light is reflected back into the medium from which it came, a phenomenon called ***total internal reflection***.



Total Internal Reflection of Light

- Total internal reflection occurs only when light travels from a higher index medium toward a lower-index medium. It does not occur when light propagates in the reverse direction—for example, from air to water.



Critical Angle θ_C: It is the angle of incidence in the medium of higher refractive index corresponding to angle of refraction of 90° in the medium of lower refractive index.

- Applying Snell's law: $n_1 \sin\theta_C = n_2 \sin 90$

$$\sin\theta_C = \frac{n_2}{n_1} \longrightarrow \theta_C = \sin^{-1}\frac{n_2}{n_1}$$

Example 3:

A beam of light is propagating through diamond $(n_1 = 2.42)$ and strikes a diamond–air interface at an angle of incidence of 28°.

a) Will part of the beam enter the air $(n_2 = 1)$ or will the beam be totally reflected at the interface?

b) Repeat part (a), assuming that the diamond is surrounded by water $(n_2 = 1.33)$ instead of air.

Solution

a)

$$\theta_C = \sin^{-1}\frac{n_2}{n_1} = \sin^{-1}\frac{1}{2.42} = 24.4°$$

Because the angle of incidence of $28°$ is greater than the critical angle, there is no refraction, and the light is totally reflected back into the diamond.

b)

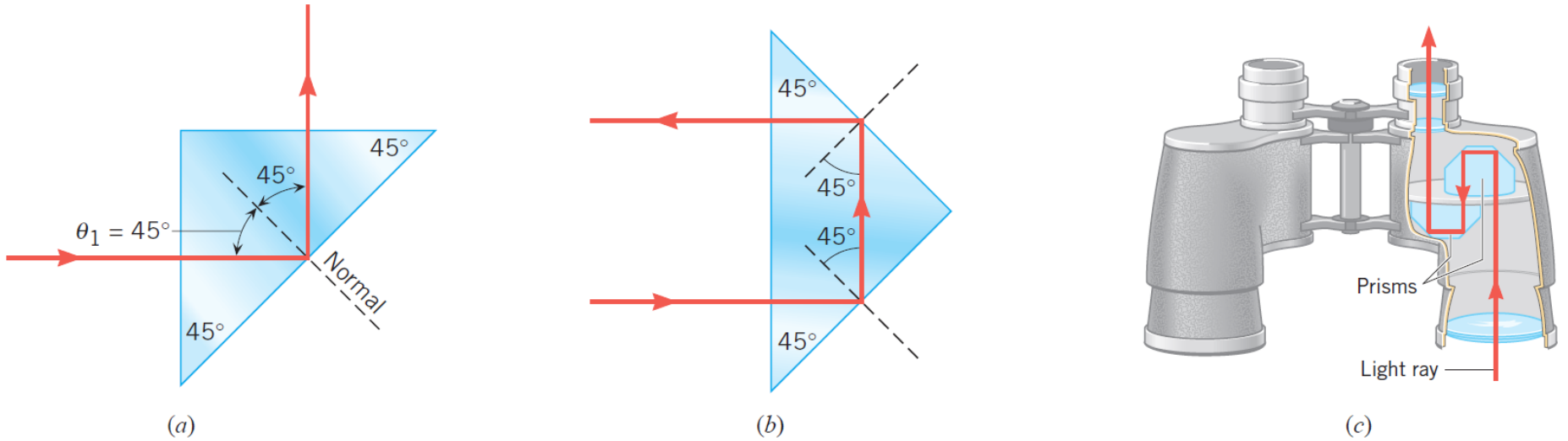
$$\theta_C = \sin^{-1}\frac{n_2}{n_1} = \sin^{-1}\frac{1.33}{2.42} = 33.3°$$

Now a beam of light that has an angle of incidence of $28°$ (less than the critical angle of $33.3°$) at the diamond–water interface is refracted into the water, so it will be partially refracted - partially reflected.

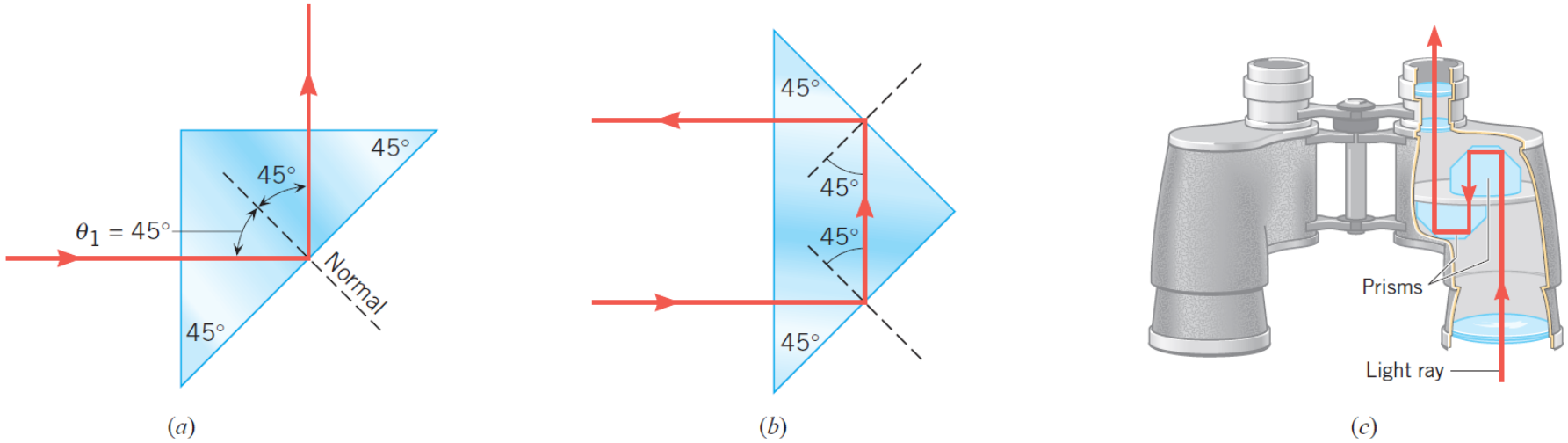
Applications of Total Internal Reflection of Light

- Many optical instruments, such as binoculars, periscopes, and telescopes, use glass prisms and total internal reflection to turn a beam of light through 90° or 180°. Figure (a) shows a light ray entering a 45°-90°-45° glass prism ($n_1 = 1.5$) and striking the hypotenuse of the prism at an angle of incidence of $\theta_1 = 45°$.
- The critical angle for a glass–air interface is

$$\theta_C = \sin^{-1}\frac{n_2}{n_1} = \sin^{-1}\frac{1}{1.5} = 42°$$

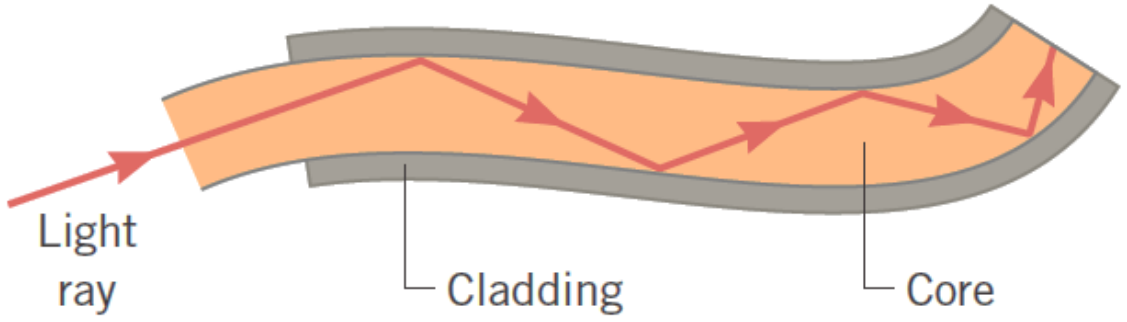


- Since the angle of incidence is greater than the critical angle, the light is totally reflected at the hypotenuse and is directed vertically upward in the drawing, having been turned through an angle of 90°.
- Part (b) of the picture shows how the same prism can turn the beam through 180° when total internal reflection occurs twice.
- Figure (c) illustrates such an application in binoculars.



Applications of Total Internal Reflection of Light

- An important application of total internal reflection occurs in fiber optics, where hair-thin threads of glass or plastic, called optical fibers, “pipe” light from one place to another.
- The figure shows that an optical fiber consists of a cylindrical inner core that carries the light and an outer concentric shell, the cladding. The core is made from transparent glass or plastic that has a relatively high index of refraction. The cladding is also made of glass, but of a type that has a relatively low index of refraction.
- Light enters one end of the core, strikes the core/cladding interface at an angle of incidence greater than the critical angle, and, therefore, is reflected back into the core. Light thus travels inside the optical fiber along a zigzag path.



- In the field of medicine, optical fiber cables have had extraordinary impact. In the practice of endoscopy, for instance, a device called an endoscope is used to peer inside the body.

Huygens' principle

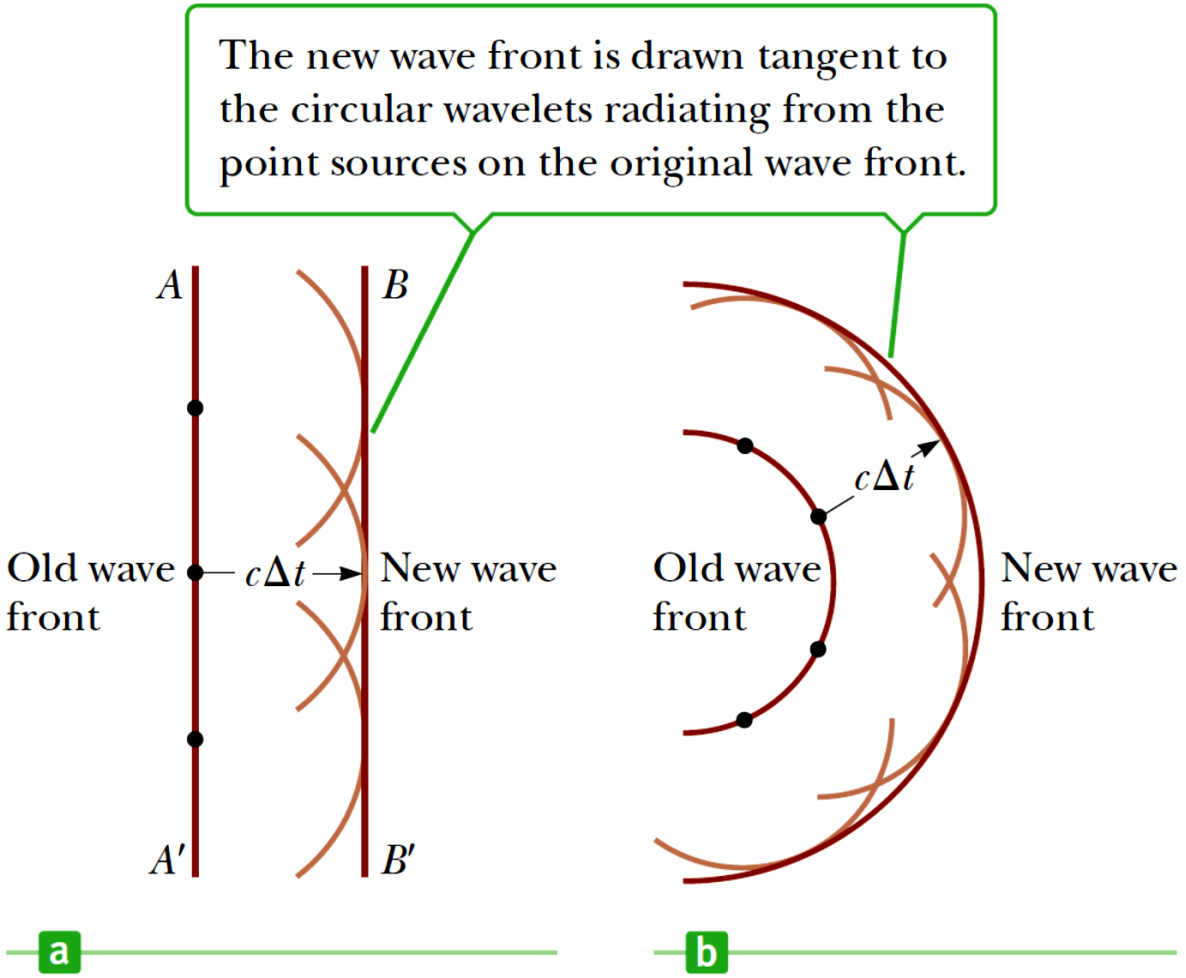
- Huygens assumed that light consists of waves rather than a stream of particles.
- A wave front is a surface passing through those points of a wave which have the same phase and amplitude.
- Huygens' principle is a geometric construction for determining at some instant the position of a new wave front from knowledge of the wave front that preceded it.
- In Huygens' construction:

All points on a given wave front are taken as point sources for the production of spherical secondary waves, called wavelets, that propagate in the forward direction with speeds characteristic of waves in that medium. After some time has elapsed, the new position of the wave front is the surface tangent to the wavelets.

Types of waves:

1) Plane wave (coming from sources at large distance “infinity, infinite radius, plane wave front, Example: Sun)

2) Spherical wave (coming from near sources, definite radius, spherical wave front, example: lamp)



Huygens’ constructions for (a) a plane wave propagating to the right and (b) a spherical wave.

Huygens' principle applied to derive the reflection and refraction law:

1- Reflection

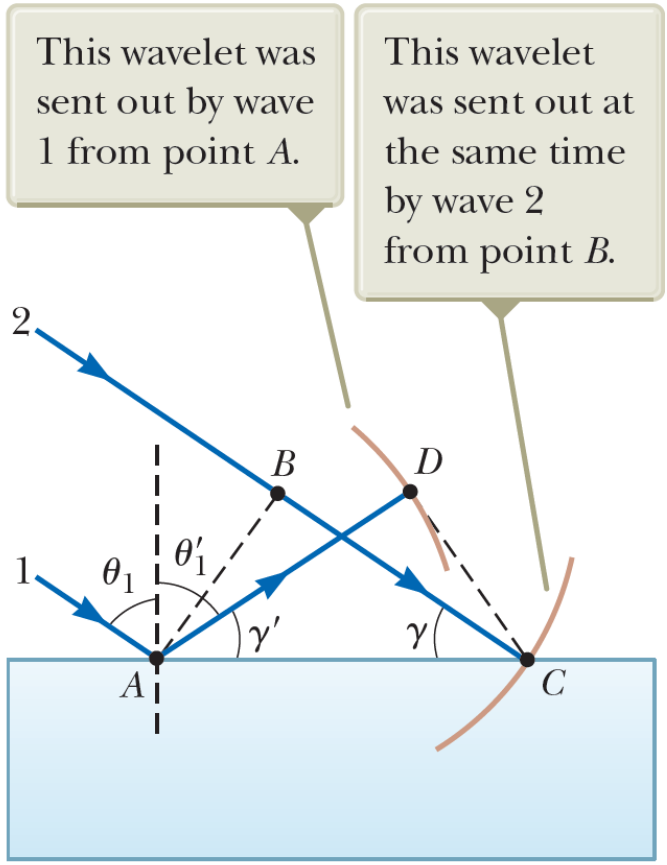
- The line AB represents a wave front of the incident light. In the same time in which ray 2 travels from B to C, ray 1 reflects from A and produces a spherical wavelet of radius AD.

$$\because BC = vt \quad \& \quad \because AD = vt \longrightarrow \therefore BC = AD$$

- Note that the right triangles ADC and ABC are congruent because they have the same hypotenuse AC and because $AD = BC$.

$$\cos\gamma = \frac{BC}{AC}, \qquad \cos\gamma' = \frac{AD}{AC} \quad \& \quad \because BC = AD$$

$$\therefore \sin\theta_1 = \sin\theta_1{}' \rightarrow \gamma = \gamma' \rightarrow \theta_1 = \theta_1{}' (Reflection\ law)$$



$$\gamma = 90 - \theta_1$$
$$\gamma' = 90 - \theta_1{}'$$

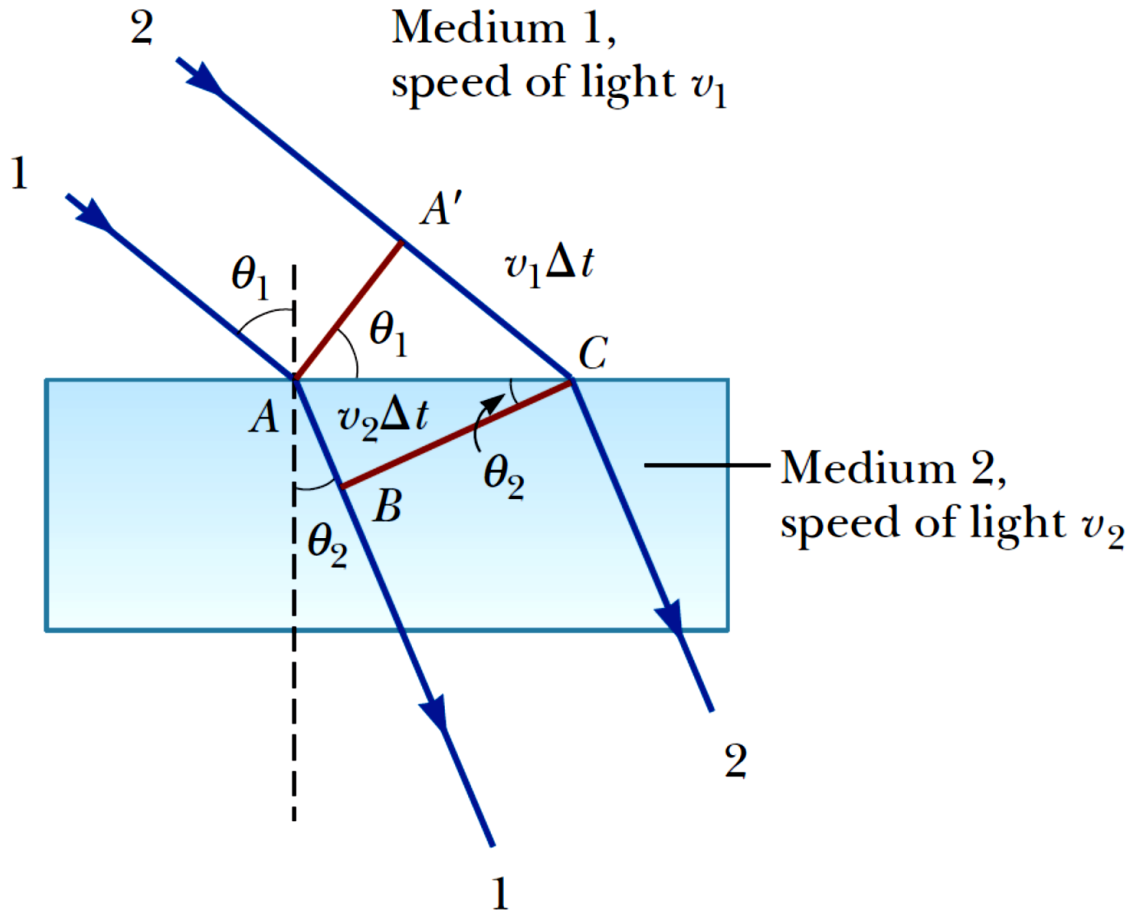
Huygens' principle applied to derive the reflection and refraction law:

2- Refraction

- In the same time interval Δt ray 1 moves from A to B in medium 2 with speed v_2 & ray 2 moves A' to C in medium 1 with speed v_1.

$$\sin\theta_1 = \frac{A'C}{AC} = \frac{v_1\Delta t}{AC} \qquad (1)$$

$$\sin\theta_2 = \frac{AB}{AC} = \frac{v_2\Delta t}{AC} \qquad (2)$$

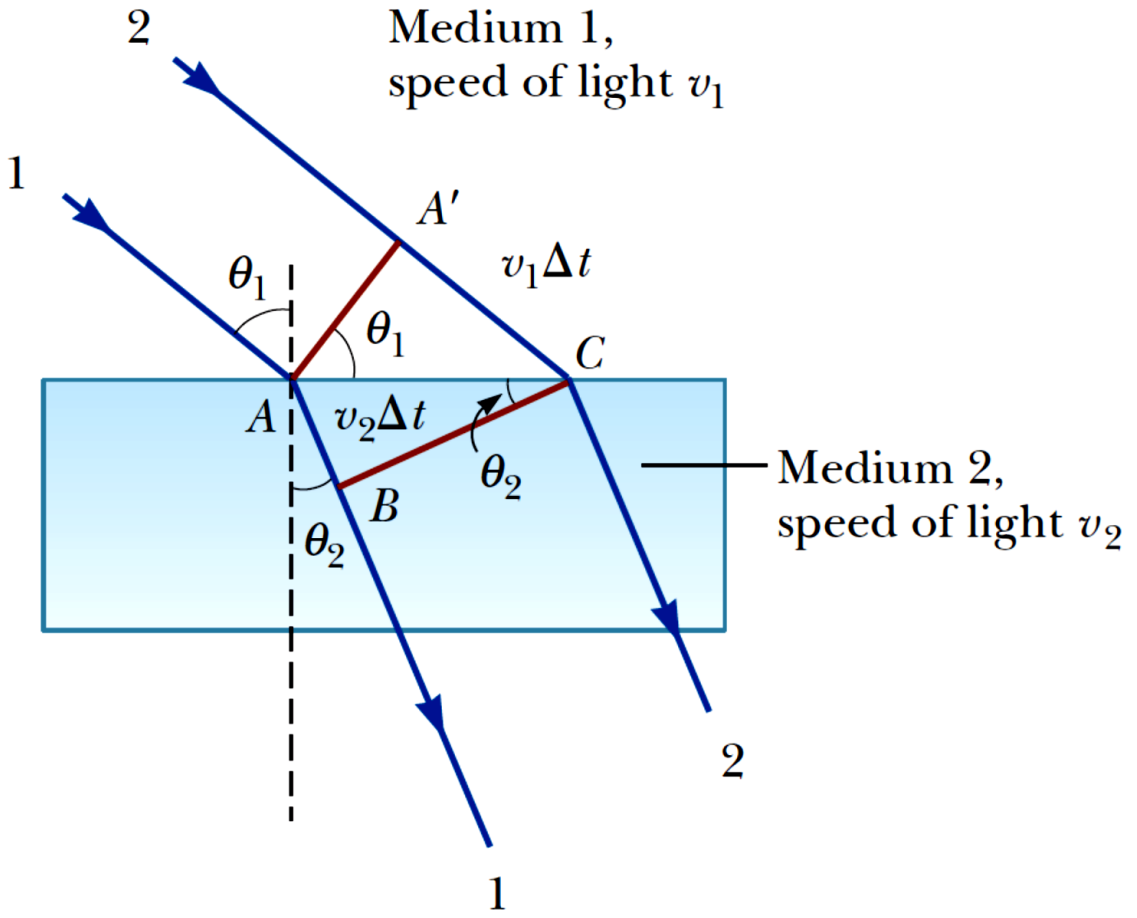


(1) divide (2):

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{\frac{v_1 \cancel{\Delta t}}{\cancel{AC}}}{\frac{v_2 \cancel{\Delta t}}{\cancel{AC}}}$$

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2} \quad (3)$$

$$\because v_1 = \frac{c}{n_1} \quad \& \quad v_1 = \frac{c}{n_2} \quad (4)$$



From (4) in (3):

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{\frac{\cancel{c}}{n_1}}{\frac{\cancel{c}}{n_2}} = \frac{n_2}{n_1}$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \quad (Snell's\ law)(Refraction\ law)$$